

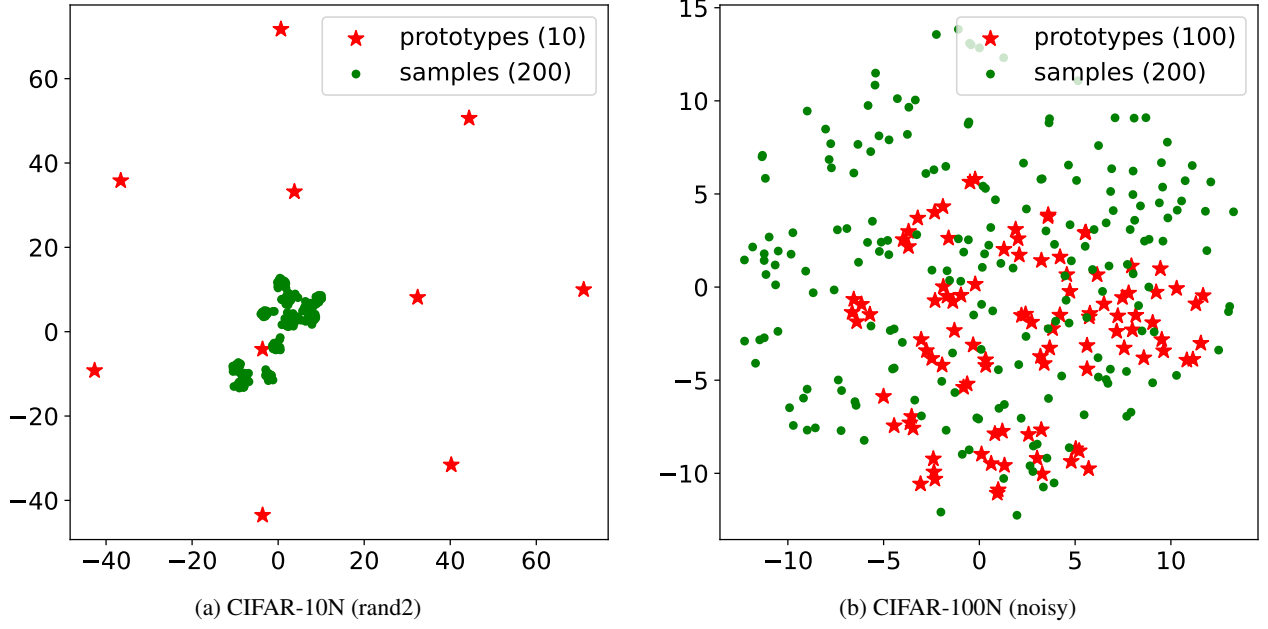


Figure 4: Visualization of prototype embedding on the CIFAR-N dataset.

Table 3: The label correction results of small circle in Proto-semi ($T_w = 20, \alpha = 0.95$).

		CIFAR-10N				CIFAR-100N
	aggre	rand1	rand2	rand3	worst	noisy
Size of unconfident dataset	10534	10995	10611	5836	21946	25096
Number of samples in small circle	4015	5766	4258	1729	9472	1823
Number of label corrections:	3992	5703	4235	1713	9381	1708
Right Number of label corrections	3846	5416	4097	1666	8723	1325
Correction accuracy	96.34%	94.97%	96.74%	97.26%	92.99%	77.58%

accuracy in label correction for all CIFAR-10N datasets compared to CIFAR-100N. These findings are consistent with the results presented in Figure 4.

Conclusion

In this paper, we present a novel sample selection-based approach, denoted as Proto-semi, for effectively identifying the type of noise and utilizing its semantics in real-world annotation scenarios. Specifically, Proto-semi divides the entire dataset into two subsets, namely confident and unconfident datasets, and subsequently leverages a distance-based technique based on prototype vectors to identify the types of annotation noise. Furthermore, the confident and unconfident datasets are adjusted based on the distances, and a semi-supervised learning strategy is incorporated to enhance the training process. Experimental results on the CIFAR-N dataset demonstrate the effectiveness of Proto-semi in addressing the problem of learning with noisy labels in real-world annotation scenarios. Avenue of future work will focus on extending our proposed method to more real-world annotation noisy datasets.

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